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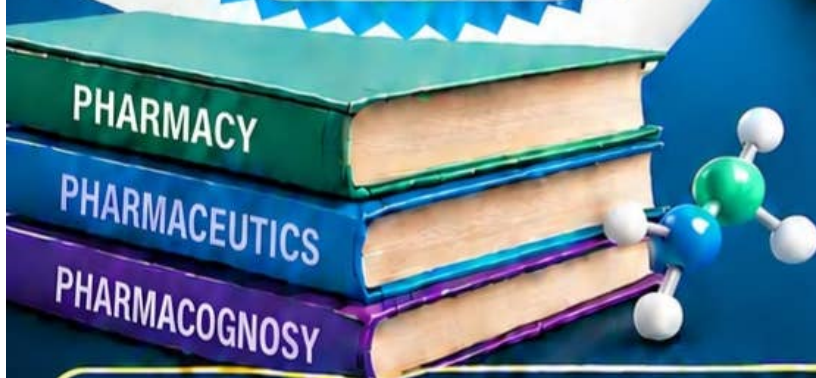
D. PHARMACY

1ST YEAR STUDENTS

**1ST
YEAR**

PHARMACOGNOSY

As per the
PCI Syllabus
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Subject wise
suggestive notes



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INDRANI PATHSALA**(Your Study Partner For Success)****(D.Pharm No.1 Digital Platform)****PHARMACOGNOSY****1st year Suggestive Note**
Session:-2025-2026**This note is applicable only
for the 2026 (1st year)
Academic Session****Content**

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CHAPTER-1**INTRODUCTION TO PHARMACOGNOSY****Summary**

Pharmacognosy is the scientific study of drug substances of natural origin—derived from plants, animals, minerals, marine sources, microorganisms and biotechnology—encompassing their physical, chemical and biological properties. The term was coined by C.A. Seydler in 1815, though the practice dates back to ancient civilizations like India (Ayurveda), Egypt (Papyrus Ebers) and Greece. Key historical figures include Dioscorides, the "Father of Pharmacognosy," and Galen, the first pharmacist. The field has evolved from simple herbal lore to the isolation of potent alkaloids like morphine and quinine. Today, it serves as a vital bridge between traditional Ayurvedic medicine and modern Allopathic systems. Its scope includes the cultivation, quality control and phytochemical analysis of natural drugs. Research now focuses on advanced areas like ethnopharmacology, phytotherapy and biotechnology to discover new therapeutic agents. Ultimately,

pharmacognosy ensures the safety and efficacy of natural products used in the pharmaceutical, food and spice industries.

Multiple Choice Questions

1. Q. The term 'Pharmacognosy' was first introduced by:
A) Hippocrates B) C.A. Seydler C) Dioscorides D) Aristotle
2. Q. In which year was the term Pharmacognosy coined?
A) 1815 B) 1806 C) 1753 D) 1600
3. Q. Who is known as the 'Father of Pharmacognosy'?
A) Galen B) Aristotle C) Dioscorides D) Theophrastus
4. Q. The word 'Pharmacognosy' is derived from which language?
A) Latin B) Greek C) French D) German
5. Q. Which of the following is an example of an organized drug?
A) Honey B) Shark liver oil C) Senna leaf D) Agar-agar
6. Q. Who is considered the 'Father of Medicine'?
A) Dioscorides B) Hippocrates C) Galen D) Sertuerner
7. Q. The first alkaloid isolated from opium in 1806 was:
A) Quinine B) Morphine C) Strychnine D) Emetine
8. Q. The study of the traditional use of plants for medicinal purposes is called:
A) Phytotherapy B) Phytochemistry C) Medical ethno botany D) Zoo pharmacognosy
9. Q. Which scientist introduced the 'binomial system' of naming plants?
A) Bentham and Hooker B) Swede Linnaeus C) Shen Nung D) Pliny the Elder
10. Q. The ancient Egyptian document containing over 800 formulae and 700 drugs is known as:
A) Charaka Samhita B) Pen-t'sao C) Papyrus Ebers D) De Materia Medica

Short Answer Questions

1. Q. Define Pharmacognosy.

Indrani Pathsala Answer: Pharmacognosy is the study of the physical, chemical, biochemical and biological properties of drugs, drug substances or potential drugs of natural origin, as well as the search for new drugs from natural sources.

2. Q. What are unorganized drugs?

Indrani Pathsala Answer: Unorganized drugs are prepared from plants but are not direct parts of plants and do not contain cellular tissues; they are prepared by processes like incision, drying or extraction.

3. Q. Name two historical Indian medicinal texts.

Ans: Charaka Samhita and Sushruta Samhita.

4. Q. What is Galenical Pharmacy?

Indrani Pathsala Answer: It is the branch of pharmacy dealing with the extraction of plant and animal drugs, named after Galen, the first pharmacist.

5. Q. What is the scope of Pharmacognosy regarding modern drug delivery?

Indrani Pathsala Answer: Its scope includes the modernization of traditional dosage forms through modern disciplines like biotechnology and novel drug delivery systems like phytosomes.

Fill in the Blanks

1. Q. The word 'Pharmakon' means _____ and 'Gignosco' means _____.

Indrani Pathsala Answer: a drug, to acquire knowledge of

2. Q. _____ isolated morphine from opium in 1806.

Indrani Pathsala Answer: Sertuerner

3. Q. The oldest known herbal, written by Shen Nung, is called _____.

Indrani Pathsala Answer: Pen-t'sao

4. Q. Pharmacognosy acts as a vital link between _____ and _____ systems of medicine.

Indrani Pathsala Answer: Ayurvedic, Allopathic

5. Q. The study of chemicals derived from plants is known as _____.

Indrani Pathsala Answer: Phytochemistry

Long Questions

1. Q. Discuss the distinguish between organized and unorganized drugs.

Indrani Pathsala Answer:

S.No.	Feature	Organized Drugs (Cellular)	Unorganized Drugs (Acellular)
1	Definition	These are direct parts of plants or animals (organs) with a definite structure.	These are products derived from plants/animals via extraction, incision or secretion.
2	Cellular Structure	They possess a well-defined cellular organization and tissues.	They do not possess cells or any anatomical organization.
3	Morphology	Can be identified by their external morphological characters (leaves, roots, etc.).	Identified by physical characters like solubility, density and refractive index.
4	Microscopy	Microscopic examination (transverse sections) is essential for identification.	Microscopy is not required; chemical and physical tests are primary.
5	Physical State	These are almost always solids.	They can be solids, semi-solids or liquids (e.g., oils, juices).
6	Preparation	Usually require only collection and drying before use.	Prepared through physical processes like incision, expression or solvent extraction.
7	Botanical Terms	Described using botanical/zoological terms (e.g., apex, margin, base).	Described by their chemical nature or physical constants.
8	Examples	Digitalis (leaves), Cinchona (bark), Clove (flower bud), Fennel (fruit).	Opium (latex), Acacia (gum), Castor Oil (fixed oil), Honey (secretion).

2. Q. Write definition, history, present status and scope of Pharmacognosy.

Indrani Pathsala Answer:

➤ Definition:

Pharmacognosy is the study of crude drugs derived from natural sources like plants, animals and minerals, including their physical, chemical and biological properties

➤ History of Pharmacognosy:

The history of pharmacognosy is as old as human civilization itself, evolving through various eras:

Ancient Period (Traditional Knowledge):

- Since ancient times, humans used plants and animals to treat ailments. In India, the **Ayurveda** system (dating back to 3500 B.C.) described the medicinal use of hundreds of plants in texts like the Charaka Samhita and Sushruta Samhita.
- Other ancient civilizations, including the Egyptians (Ebers Papyrus), Chinese (Pen T'sao) and Greeks, also maintained extensive records of natural medicines.

Modern Development (19th Century to Present):

- The 19th century marked the shift from crude drugs to the isolation of active constituents, such as the discovery of Morphine from Opium and Quinine from Cinchona.
- Today, it has evolved into a sophisticated science that uses modern techniques like **Spectroscopy** and **Chromatography** to study the chemical constituents and therapeutic efficacy of natural products.

➤ Present Status of Pharmacognosy

Currently, Pharmacognosy is considered a **multidisciplinary science** that bridges traditional knowledge with modern technology. Its present status is defined by the following key areas:

- **Rise of Phytopharmaceuticals**

In countries like India, the government has recognized **Phytopharmaceuticals** as a separate class of drugs. These are purified, standardized fractions of plant extracts that must pass rigorous clinical trials and safety tests, just like modern allopathic medicines.

- **Integration with Global Healthcare**

With the "Herbal Wave" sweeping the world, over **60-80% of the global population** relies on natural products. Pharmacognosy now provides the scientific evidence for the **AYUSH** systems (Ayurveda, Unani, etc.) and is a primary source for "Lead Compounds" in the discovery of new drugs for cancer, malaria and diabetes.

➤ Scope of Pharmacognosy:

Pharmacognosy is a vital link between traditional medicine and modern pharmaceutical science. Its scope includes:

- **Bridge Between Sciences:** It acts as a bridge between pharmacology, medicinal chemistry and pharmaceuticals. It provides the base knowledge for drug discovery from natural sources.
- **Standardization and Quality Control:** It provides methods to identify, authenticate and detect **adulteration** in crude drugs using morphological, microscopic and chemical evaluation.
- **Biotechnology (Plant Tissue Culture):** Advanced pharmacognosy uses tissue culture to produce secondary metabolites in a controlled laboratory environment, reducing the need for wild harvesting.

CHAPTER-2

CLASSIFICATION OF CRUDE DRUGS

Summary

Crude drugs from natural sources are systematically classified using six primary methods to facilitate scientific study and identification. The **Alphabetical** system organizes drugs by their names in various languages or pharmacopoeias, offering simplicity but lacking scientific depth. **Taxonomical** classification relies on biological hierarchy (phylum to species), providing evolutionary insights despite ignoring chemical or medicinal activities. **Morphological** classification divides drugs into **Organized** (cellular tissues like leaves or roots) and **Unorganized** (acellular substances like gums or resins) categories, which is highly practical for identifying physical adulterants. **Chemical** classification groups drugs by active constituents such as alkaloids or glycosides, making it ideal for phytochemical research. **Pharmacological** classification arranges drugs based on their therapeutic effects, such as cardiotonics or laxatives, regardless of their botanical origin. Finally, **Chemotaxonomical** classification—the most modern approach—links chemical similarity with taxonomic status to understand plant evolution and biosynthetic relationships.

Important Abbreviation

USP	United States Pharmacopoeia
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Multiple Choice Questions

1. Q. Which classification system is used in the Indian Pharmacopoeia (IP)?

- A) Morphological B) Alphabetical C) Chemical D) Pharmacological

2. Q. Crude drugs are arranged according to their botanical natural identity in:

- A) Taxonomical classification B) Chemical classification
C) Alphabetical classification D) Morphological classification

3. Q. In morphological classification, "Shark Liver Oil" is classified as:

- A) Organized drug B) Vegetative drug C) Unorganized drug D) Cellular drug

4. Q. Which system classifies drugs based on their therapeutic use?

- A) Morphological B) Taxonomical C) Pharmacological D) Chemical

Ans: C) Pharmacological

5. Q. "Tropane alkaloids in Solanaceae family" is an example of which classification?

- A) Alphabetical B) Chemotaxonomical C) Morphological D) Pharmacological

6. Q. Which of the following is an advantage of Morphological classification?

- A) It shows the chemical nature B) It helps identify adulterants
C) It shows evolutionary history D) It is based on therapeutic action

7. Q. Drugs like Senna, Digitalis and Cinchona are classified as:

- A) Organized drugs B) Unorganized drugs C) Mineral drugs D) Synthetic drugs

8. Q. A disadvantage of Pharmacological classification is:

- A) It is easy to study B) It ignores the chemical constituents
C) Drugs with different actions are grouped together
D) It doesn't provide morphological or taxonomical info

9. Q. In which classification do we use 'Phylum, Class, Order, Family'?

- A) Chemical B) Alphabetical C) Taxonomical D) Morphological

10. Q. The classification based on the presence of active constituents like alkaloids or glycosides is:

- A) Chemical classification B) Morphological classification
C) Taxonomical classification D) Pharmacological classification

Short Answer Questions

1. Q. What is Alphabetical Classification?

Indrani Pathsala Answer: It is a system where crude drugs are arranged in alphabetical order of their Latin, English or local names, commonly used in Pharmacopoeias.

2. Q. Define 'Organized Drugs'.

Indrani Pathsala Answer: Organized drugs are crude drugs that consist of cellular tissues and represent direct parts of the plant, such as leaves, roots or barks.

3. Q. Give one example of Pharmacological Classification.

Indrani Pathsala Answer: Purgatives (e.g., Senna, Castor oil) or Cardiotonics (e.g., Digitalis).

4. Q. What is Chemotaxonomical Classification?

Indrani Pathsala Answer: It is a modern classification system that combines chemical facts (constituents) with the taxonomic position (botanical status) of the plant.

5. Q. Name two reference books that use Alphabetical Classification.

Indrani Pathsala Answer: 1. Indian Pharmacopoeia (IP), 2. British Pharmacopoeia (BP).

Fill in the Blanks

1. Q. _____ drugs are prepared from plants by physical processes like incision, drying or extraction and do not contain cellular tissue.

Indrani Pathsala Answer: Unorganized

2. Q. In _____ classification, drugs are arranged according to Phylum, Division, Class, Order and Family.

Indrani Pathsala Answer: Taxonomical

3. Q. _____ classification is very useful for identifying the presence of adulterants in crude drugs.

Indrani Pathsala Answer: Morphological

4. Q. Drugs like _____ and _____ are examples of organized drugs because they contain cellular structure.

Indrani Pathsala Answer: Senna, Cinchona

5. Q. The _____ classification is considered the most preferred method for the study of phytochemistry and pharmacological activity.

Indrani Pathsala Answer: Chemical

Long Questions

1. Q. Explain the different systems of classification of Crude Drugs with examples.

Indrani Pathsala Answer:

➤ Definition of Crude Drug

A Crude Drug is defined as a natural product—derived from plants, animals or minerals—that has not been advanced in value or improved in condition by any process beyond what is essential for its collection and drying.

Key Characteristics:

- They exist in their raw, unrefined form.
- Their molecular structure remains unchanged from how it is found in nature.
- They are used for the diagnosis, cure, mitigation, treatment or prevention of disease.

Classification of Crude Drugs

Since crude drugs are obtained from various sources (plants, animals, minerals), it is essential to classify them systematically for easy study.

a. Morphological Classification:

This system groups crude drugs based on their external (morphological) characters or the specific part of the plant/animal from which they are obtained. It is primarily divided into two categories

- **Organized Drugs (Cellular)**

These are the direct parts of plants or animals that possess a definite cellular structure.

Leaves: Senna, Digitalis, Tulsi.

Barks: Cinnamon, Cinchona.

Flowers: Clove, Saffron.

Roots/Rhizomes: Rauwolfia, Ginger, Turmeric

- **Unorganized Drugs (Acellular)**

These are products derived from natural sources through physical processes (like extraction or incision) and lack a cellular structure.

Dried Latex: Opium.

Dried Juice: Aloe.

Gums: Acacia, Tragacanth.

Volatile Oils: Clove oil, Peppermint oil.

b. Taxonomical Classification:

It is a purely biological system of classification where crude drugs are arranged according to the botanical (plants) or zoological (animals) kingdom. It is based on the evolutionary relationship between organisms.

Examples: Family Umbelliferae: Fennel, Coriander, Caraway.

The Hierarchy:

The drugs are grouped into a specific order:

Kingdom -> Phylum -> Class -> Order -> Family -> Genus -> Species

Family Solanaceae: Belladonna, Datura, Ashwagandha.

c. Chemical Classification:

Drugs are classified based on the active chemical constituents present in them, which are responsible for their medicinal action.

Examples:

Alkaloids: Cinchona, Ipecac, Tea.

Glycosides: Digitalis, Senna, Aloe.

Volatile Oils: Clove, Eucalyptus, Peppermint.

Lipids: Castor oil, Cod liver oil.

d. Pharmacological (Therapeutic) Classification:

Drugs are grouped according to their medicinal action or their effect on the human body systems.

Examples:

Purgatives: Senna, Castor oil.

Cardiotonics: Digitalis, Arjuna.

Carminatives: Fennel, Cardamom.

Antihypertensives: Rauwolfia.

e. Alphabetical Classification

In this system, crude drugs are arranged in the alphabetical order of their **English names** or **Latin names**.

Some pharmacopoeias also use local vernacular names for this purpose.

Example: The drugs are listed as:

A – Acacia, Aloe, Amla

B – Benzoin, Beeswax, Belladonna

C – Cinchona, Clove, Coriander

D – Digitalis, Datura

Usage: Adopted by Pharmacopoeias (IP, BP, USP).

f. Chemotaxonomical Classification:

Chemotaxonomical classification is a modern system of arranging crude drugs based on the presence of **specific chemical constituents** within **specific taxonomic groups** (families, genera, etc.). It combines the principles of **Chemistry** and **Taxonomy**

Example: Presence of Tropane alkaloids in the family Solanaceae.

2. Q. Define and explain the Morphological classification and Pharmacological classification of Crude drugs with examples. Differentiate the two classification of Crude drugs with examples.

Indrani Pathsala Answer:

➤ Morphological Classification:

This system groups crude drugs according to the **plant or animal part** used as the medicine. It is primarily used to distinguish between organized and unorganized drugs.

- **Organized Drugs (Cellular)**

These are direct parts of plants or animals consisting of cellular tissues.

Leaves: Digitalis, Senna, Tulsi.

Barks: Cinchona, Cinnamon, Arjuna.

Flowers: Clove, Saffron.

Fruits: Fennel, Coriander, Amla.

Seeds: Nux-vomica, Isabgol.

Roots/Rhizomes: Rauwolfia, Ginger, Turmeric.

- **Unorganized Drugs (Acellular)**

These are derived from plants/animals via physical processes (extraction/secretion) and lack cellular structure.

Dried Latex: Opium, Papain.

Dried Juices: Aloe, Kino.

Gums: Acacia, Tragacanth.

Resins: Benzoin, Asafoetida.

Fixed Oils/Waxes: Castor oil, Beeswax.

➤ Pharmacological Classification:

Pharmacological classification (also known as **Therapeutic classification**) is a system where crude drugs are grouped according to their **medicinal action**, therapeutic uses or their effects on various systems of the human body, also known as **Therapeutic Classification**

Examples by Action:

- **Gastrointestinal Tract (GIT):**

Laxatives/Purgatives: Senna, Castor oil, Isabgol.

Carminatives: Fennel, Cardamom, Ginger.

- **Cardiovascular System (CVS):**

Cardiotonics: Digitalis, Arjuna.

Anti-hypertensives: Rauwolfia.

- **Respiratory System:**

Expectorants: Vasaka, Liquorice.

Anti-tussives: Opium (Codeine).

- **Central Nervous System (CNS):**

CNS Stimulants: Coffee, Tea.

CNS Depressants: Opium.

➤ Comparison Table: Morphological vs. Pharmacological:

S.No.	Feature	Morphological Classification	Pharmacological Classification
1	Basis of Grouping	Drugs are grouped based on the part of the plant used (e.g., leaves, barks, roots).	Drugs are grouped based on their medicinal action or therapeutic use.
2	Primary Focus	Focuses on the physical structure and appearance of the drug.	Focuses on the effect on the human body systems (e.g., CVS, GIT).
3	Identification	Very helpful in identifying raw drugs in the lab using the naked eye.	Not useful for identification; only explains what the drug does.
4	Sub-categories	Divided into Organized (cellular) and Unorganized (acellular) drugs.	Divided by Body Systems (e.g., Carminatives, Purgatives, Cardiotonics).
5	Adulteration	Highly effective in detecting adulteration by observing physical changes.	Not useful for detecting adulteration or impurity.
6	Consistency	A drug usually belongs to one group (e.g., Senna is always a "leaf").	A drug may belong to multiple groups (e.g., Cinchona is a bitter and an antimalarial).
7	Utility	Most useful for Pharmacognosists and laboratory technicians.	Most useful for Pharmacists and Doctors for clinical practice.
8	Examples	Leaves: Senna, Digitalis. Bark: Cinchona, Cinnamon.	Purgatives: Senna, Castor Oil. Cardiotonics: Digitalis, Arjuna.